

* 3 Types of Machine Learning.

Machine Learning Techniques.

Supervised ML

Unsupervised ML

Reinforcement Learning.

#1. Supervised ML Techniques:

Regression
Classification

Predict House Price.

Dataset → Independent features → Dependent feature or O/P feature

Size of House	No. of Rooms	Price.
5000	5	450K
6000	6	500K
—	—	—
—	—	—

Continuous

(1) Dependent or output feature

(2) Continuous → Regression

(3) Categorical → classification.

Classification.

No. of study hours	No. of play hours	Pass / Fail
7	3	Pass
2	6	Fail
—	—	May Be

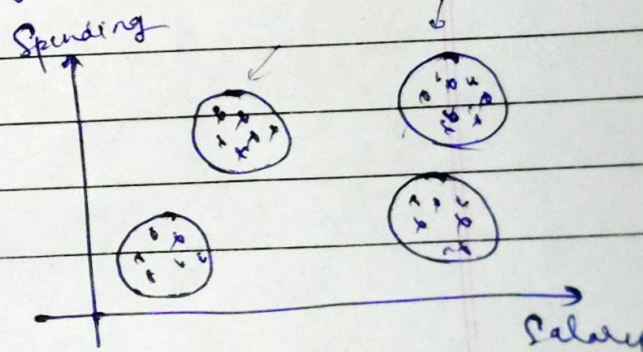
Dependent feature

Binary Classification → Only 2 categories

Multiclass classification → more than 2 categories.

#2. Unsupervised ML + Eg + Customer Segmentation. ⇒ Clusters

Salary	Spending score (1-10)
20000	9
45000	2
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Supervised ML

- i. Linear Regression
 - ii. Ridge & Lasso
 - iii. Elastic Net
 - iv. Logistic Regression (Classification)
 - v. Decision Tree
 - vi. Random Forest
 - vii. Ada Boost
 - viii. Xg boost
- Both
Classification
↓
Regression

Unsupervised ML

- i. K Means
- ii. Hierarchical Means
- iii. DB Scan clustering

#3. Reinforcement Learning.

Works on Reward and Penalty.